Ming Hsiao

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EDUCATION

Master 2 Research and Innovation (M2RI) *Supervisor: Professor Vincent Guedj* Sep 2026 - Jun 2027

Université de Toulouse (Université Paul Sabatier)

Master of Science (M.S.) in Mathematics *Supervisor: Professor Yng-Ing Lee* Sep 2023 - Jun 2025

National Taiwan University

Bachelor of Science (B.S.) in Mathematics Sep 2019 - Jun 2023

National Taiwan University

PAPER

1 **(with Pak-Yeung Chan) Curvature estimates for steady and expanding solitons in higher dimensions** , to appear in *Trans. AMS*. [arXiv]

2 **(with Man-Chun Lee) Gap Theorem on locally conformally flat manifold** , to appear in *Comm. Anal. Geom.* [arXiv]

3 **Rotationally symmetric Ricci Flow on $\mathbb{R}^{n+1}$,** *J. Geom. Anal.* **36** (2026), Article 66. [arXiv] [Journal]

PREPRINT

1 **Steady soliton with $\mathcal{L}^1$ decay curvature, arXiv.** [arXiv]

2 **(with Hao-Wei Huang) Limiting Spectral Distributions of Hadamard Products of GUE matrices, submitted.**

TALK AND POSTERS

PIMS-NCTS Workshop in Geometric Analysis | PIMS | *Invited speaker* Jun 2026

Rotationally symmetric Ricci flow

28th Taiwan Geometry Symposium | NCTS | *Short Communications* Nov 2025

Curvature estimates for steady and expanding solitons in higher dimensions

NCTS Differential Geometry Seminar | NCTS | *Invited speaker* Apr 2025

Gap Theorem on Locally Conformally Flat Manifold

Graduate Student Seminar | SLMath Institute | *Invited speaker* Oct 2024

Preservation of Killing Vector Fields under Ricci Flow

TMS Annual Meeting | NCCU | *Special session speaker* Jan 2024

Analysis and Optimization-Hadamard product of random matrices and their limiting spectral distributions

2023 NCTS-Kyoto Mathematics Symposium | NCTS | *Poster presentation* Dec 2023

2023 NCTS-Kyoto Mathematics Symposium

2023 Symposium for Young Analysts | NCTS | *Invited speaker* Jun 2023

Limit Theorem of Hadamard Product Random Matrices

ACADEMIC EXPERIENCE

NCTS | *Research Assistant - Mentor: Prof. Yng-Ing Lee* Aug 2025 - Dec 2025

CUHK | *Visiting - Visiting Professor: Man-Chun Lee* Dec 2024

SLMath | *New Frontiers in Curvature: Flows, General Relativity, Minimal Submanifolds, and Symmetry* Aug 2024 - Nov 2024

- NCTS Research Abroad Program - Mentor: Professor Peter Topping

SMS | *Summer school* Jun 2024

- Flows and Variational Methods in Riemannian and Complex Geometry: Classical and Modern Methods

NTU | *Graduate Research Program* Aug 2023 - Present

- Topic: Ricci Flow on Riemannian Manifolds/ Advisor: Prof. Yng-Ing Lee

64th International Mathematical Olympiad | *2023 IMO Taiwan team Observer A* Jul 2023

National Science and Technology Council | *National Science Council Research Scholarship, TW* Mar 2022 - Mar 2023

- Topic: Free Probability and Hadamard Random Matrix/ Advisor: Prof. Hao-Wei Huang

NCTS | *Undergraduate Research Program* Oct 2021 - Jun 2022

- Topic: Free Probability and Hadamard Random Matrix/ Advisor: Prof. Hao-Wei Huang

NCTS | *Undergraduate Summer Research Program* Jul 2021 - Aug 2021

- Topic: Free Probability and Hadamard Random Matrix/ Advisor: Prof. Hao-Wei Huang